Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026
Technical & Tools Declaration
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1. Team & Project Overview

Team Name	Here We Go
Project / Solution Name	IkiGuy.
Name (Leader)	Abel Nathanael Hutapea
Email & Phone (Leader)	DSC2309294@xmu.edu.my & +60-147104970
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://here-we-go-sdc.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/ProProcer/here-we-go-sdc

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.
[Maximum 200 words]

FutureReady AI connects a React and TypeScript interface to a Python FastAPI backend through REST APIs. Users upload a synthetic CV or enter a profile. The backend extracts document text using PDF, Word and OCR tools, normalizes recognized skills against the O*NET catalog, and requests AI-assisted proficiency estimates for user review.

AI requests use Groq's GPT-OSS-120B as the primary model, with RunPod-hosted GLM-4.7-Flash and Google Gemini as fallbacks. Validated results return to the interface for confirmation. The recommendation engine then calculates explainable career matches and skill gaps using deterministic scoring.

Versioned JSON catalogs store occupations, skills, dated research evidence, verified courses and SDG mappings. Temporary profiles remain in expiring backend memory. Research evidence informs career outlooks and clearly labelled hypothetical scenarios, while course and SDG catalogs support personalized learning roadmaps.

Replit hosts the built frontend and FastAPI backend together, providing one public application URL. RunPod separately hosts the GPU inference endpoint.

The end-to-end flow is: User → React interface → FastAPI backend ↔ AI providers and structured catalogs → career matches, scenarios and learning roadmap → user interface, with Replit hosting the application.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Core backend development language	PSF License
TypeScript 7.0.2 / JavaScript (ECMAScript)	Frontend logic and development	Apache 2.0 / Standard
HTML5 / CSS3	Frontend structure and styling	W3C Standard
React / React DOM 19.3.1	Frontend UI components and rendering	MIT
Vite 8.3.5	Frontend bundling and build tooling	MIT
Tailwind CSS 4.3.3	Frontend utility styling	MIT
FastAPI 0.135.1	Backend REST API routing and logic	MIT
Pydantic 2.12.3	Backend data validation	MIT
Uvicorn 0.41.0	Backend ASGI application server	BSD-3-Clause
Pypdf 6.21.1	Document processing: PDF text extraction	BSD-3-Clause
python-docx 1.2.0	Document processing: Word documents	MIT
pypdfium2 5.5.0	Document processing: PDF rendering	Apache 2.0 / BSD
Pillow 13.0.0	Image processing	HPND
RapidOCR 3.10.0	Optical character recognition for scanned docs	Apache 2.0

Name & Version	Purpose in Project	License
ONNX Runtime 1.24.3	Execution engine for OCR models	MIT
Node.js 22.x / npm (bundled version)	Development environment and package management	MIT / Artistic 2.0
unittest (Python 3.12 standard library)	Backend testing	PSF License
Playwright 1.58.0	Automated application testing	Apache 2.0
FutureReady AI 0.2.0	Application-specific interface, career-matching logic, evidence workflows, learning roadmaps, and integration	Apache 2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Groq API	Skill-proficiency estimation, research interpretation, and evidence-review tasks (using GPT-OSS-120B)	Free
Serverless Inference API (Runpod)	Fallback model access (GLM-4.7-Flash)	Sponsor-provided
Gemini API (Google)	Final backup inference access	Free
Management APIs (RunPod)	Configure endpoints and check service status via deployment and diagnostic scripts	Sponsor-provided

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google & Microsoft Research RSS Feeds	Research discovery and retrieval of associated public articles	Free
Crossref REST API (Crossref)	Query publication metadata and supporting research for the evidence-review workflow	Free
arXiv & Publisher APIs/Pages	Retrieve relevant documents and web pages for evidence extraction	Free
Public Reports Fetching (e.g., Anthropic)	Retrieve approved public research reports (e.g., Anthropic Economic Index)	Free
GitHub REST API (GitHub)	Retrieve review-workflow artifacts and configure repository secrets and variables	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Application hosting (Single Reserved VM serving the React frontend and FastAPI backend via one public URL)	Core/-
RunPod	GPU compute (Serverless endpoint hosting GLM-4.7-Flash for fallback AI inference using an A100 80 GB GPU)	Usage-based serverless compute / US Midwest (MW04)
GroqCloud	Managed AI inference (Primary model inference)	Free/ global inference cloud
Google Gemini API	Managed AI inference (Final fallback model inference)	Free/-

Service & Provider	Purpose in Project	Plan / Region
GitHub & GitHub Actions	Repository hosting and automation (Store source code and run scheduled evidence-discovery workflows)	-/-

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-OSS-120B (openai/gpt-oss-120b)	OpenAI / Groq	Pre-trained (Primary)	Apache 2.0, GPT-OSS usage policy, Groq service terms
GLM-4.7-Flash (zai-org/GLM-4.7-Flash)	Z.ai / RunPod	Pre-trained (Fallback)	MIT (weights), RunPod service terms
Gemini 3.1 Flash-Lite (gemini-3.1-flash-lite)	Google (Gemini API)	Pre-trained (Final Fallback)	Proprietary, Google APIs and Gemini API terms
RapidOCR Models (RapidOCR 3.10.0 / ONNX Runtime)	RapidOCR	Pre-trained (Image/PDF text extraction)	Apache 2.0, applicable model attribution

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0	O*NET	Public	CC BY 4.0	N
SOC occupational classification	BLS	Public	U.S. government information; BLS attribution	N
Research articles and feed snapshots	Google Research / Microsoft Research	Public	Publisher copyright and applicable page terms	Y
Anthropic Economic Index findings	Anthropic	Public	Publisher copyright and applicable terms	N
Research publication metadata	Crossref	Public	Mostly CC0; abstracts retain source licenses	Y
Verified learning-resource catalog	OpenLearn	Self-collected from public sources	Provider terms; OpenLearn generally CC BY-NC-SA, with exceptions	N
SDG targets and role/skill mappings	UN SDGs / team mappings	Public / self-collected	UN terms; project license for original mappings	N
Fictional CVs and evaluation profiles	Project fixtures	Synthetic	Apache 2.0	N
Mock role catalogs and hypothetical scenarios	Project fixtures	Synthetic	Apache 2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Derived evidence assessments	Project review records	Self-collected / synthetic interpretations	Project license; excerpts retain source rights	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex	Code generation, debugging, integration, test development, architecture guidance and documentation drafting	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code and text were checked against project requirements, source documents and application behaviour. Verification included automated backend tests, frontend builds, browser checks, synthetic CV evaluations and live API checks. Structured validation checks AI responses against recognized skill IDs and supporting evidence. Unsupported research interpretations remain separate from approved findings. Our team is responsible for reviewing, correcting and approving the submitted implementation and documentation.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud development and hosting platform	Application import, build configuration, preview and deployment preparation for the React/FastAPI prototype
RunPod	Serverless GPU platform	Hosting GLM-4.7-Flash on an A100 GPU endpoint, with verified inference for fictional CV assessment

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:
[Maximum 100 words]

These platforms are replaceable. Replit hosts the application, while RunPod supplies fallback AI inference. The application could move to another compatible Python hosting platform, and another model endpoint could replace RunPod. Core deterministic matching remains available without AI inference. Declare sponsor-issued credits separately only if the team actually received and used them.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State “No user data is collected” if applicable.
[Maximum 150 words]

The prototype processes uploaded CV content, extracted skill evidence, proficiency estimates and user corrections. Demonstrations and recorded evaluations use fictional profiles. Uploads are parsed locally on the application server; CV files and profiles are not intentionally persisted to disk. Profiles remain in server memory, expire after one hour, and are cleared on restart. Expired entries are removed during access or session creation.

AI assessment sends selected CV evidence snippets and skill metadata to Groq, with RunPod and Gemini used as fallbacks. Snippets could contain personal information if a real CV is

uploaded; automatic anonymization is not guaranteed. Provider-side retention follows the applicable service terms. API credentials remain in environment variables or secret storage. Random session tokens restrict profile access, and API responses disable caching. Public deployment uses HTTPS.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented. [Maximum 150 words]

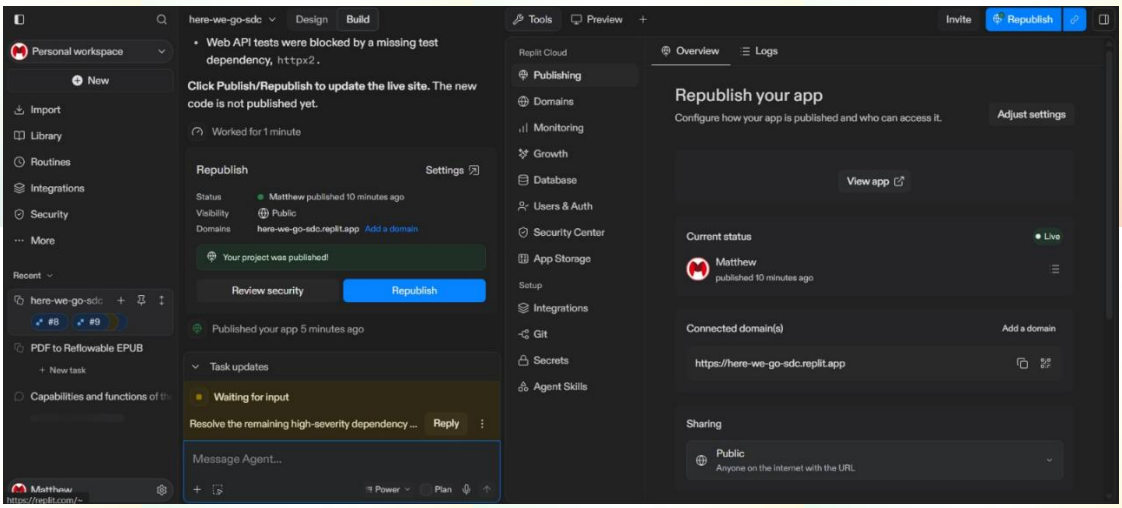
AI models may hallucinate, misinterpret CV evidence or estimate proficiency incorrectly. We constrain outputs to recognized skill IDs, validate response structure and evidence references, and allow users to review and correct assessments. Hypothetical scenarios are clearly labelled, and generated research interpretations remain separate from approved findings.

Models and O*NET's U.S.-focused occupational data may introduce bias. Recommendations therefore use transparent scoring and do not claim to predict Malaysian employment demand or guarantee jobs.

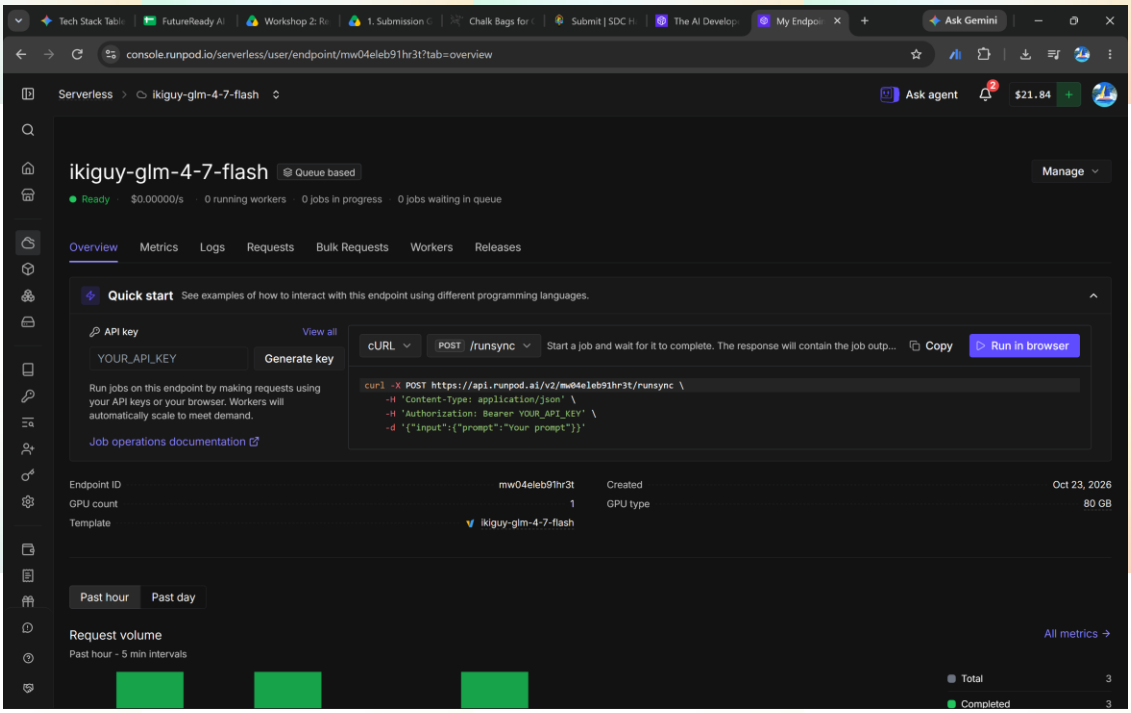
Rate limits, outages and GPU cold starts may delay AI responses. Each assessment can attempt Groq, RunPod and Gemini in sequence when availability failures occur, subject to bounded timeouts. Invalid outputs fail validation instead of triggering fallback. Deterministic matching and cached resources remain available without AI.

OCR may miss skills in complex documents, requiring user correction. Research and course coverage remain limited, with unsupported results explicitly identified.

6.3 Proofs of Use for Replit



6.4 Proofs of Use for Runpod



URL Endpoint: https://api.runpod.ai/v2/mw04eleb91hr3t

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24/10/2026